

CHAPTER-I

INTRODUCTION

Agriculture remains the backbone of India's economy, serving as the primary source of livelihood for millions of rural households. With the increasing challenges of climate change, resource scarcity, and population growth, the stability of agricultural production has become a national priority. Among various cereal crops, Pearl Millet (*Pennisetum glaucum*) — locally known as *Bajra* — stands out as one of the most resilient and climate-adaptable grains grown in India's semi-arid and drought-prone regions.

Pearl millet is often recognized as a “climate-smart” crop because of its exceptional tolerance to heat, drought, and low soil fertility. It thrives in conditions where other major crops such as rice, wheat, and maize often fail, making it crucial for ensuring food and nutritional security. It is rich in proteins, fiber, iron, and essential micronutrients, playing a vital role in combating malnutrition and promoting sustainable agriculture.

Despite its adaptability, the yield of pearl millet is not consistent, as it is influenced by multiple environmental and climatic factors such as temperature, rainfall, humidity, wind speed, and soil moisture. These parameters interact in complex, nonlinear ways that are difficult to analyze through conventional statistical techniques. Predicting the yield of pearl millet under varying climate conditions is therefore essential for supporting effective crop management, planning, and policymaking.

With the emergence of Artificial Intelligence (AI) and Machine Learning (ML), agriculture has entered a new era of data-driven decision-making. These technologies can process large datasets, uncover hidden relationships, and forecast agricultural outcomes with higher accuracy. In particular, unsupervised learning algorithms play a significant role in understanding complex, unlabeled agricultural data. They can automatically detect clusters, relationships, and trends among climatic parameters that affect crop yield.

In this project, three unsupervised learning techniques - Principal Component Analysis (PCA), Density - Based Spatial Clustering of Applications with Noise (DBSCAN), and Self-Organizing Maps (SOM) - are applied to analyze and predict the yield of pearl millet. The

study compares their effectiveness in identifying key climatic variables that influence production and in estimating yield trends under dynamic climate scenarios.

The research utilizes real-world data obtained from the Tamil Nadu Government Agriculture Department, which includes information on area, production, yield, temperature, humidity, wind speed, rainfall, and soil moisture from 2020 to 2024, with 2025 used for prediction. The dataset is processed and analyzed using the Orange AI Tool, a visual machine learning platform that simplifies model construction and performance evaluation without requiring complex programming

1.1 Background of Pearl Millet

Pearl Millet (*Pennisetum glaucum*) is one of the oldest cultivated cereal grains, belonging to the grass family Poaceae. It has been a staple food for millions of people living in the arid and semi-arid regions of India and Africa for thousands of years. Among all millets, pearl millet is the most widely grown due to its exceptional adaptability to poor soils, high temperatures, and limited rainfall. It can thrive in areas receiving as little as 250–400 mm of annual rainfall and in temperatures up to 45°C, where other major cereal crops like rice, wheat, or maize cannot survive.

In Tamil Nadu, pearl millet is mainly cultivated in Erode, Salem, Coimbatore, Dharmapuri, Villupuram, Namakkal, and Krishnagiri districts, where rainfall is moderate and soil conditions favor dryland farming. These regions are part of the state's Northwestern Agro-Climatic Zone, which experiences dry weather and limited irrigation. The crop's deep-rooted nature makes it well-suited for these areas, where it thrives on residual soil moisture and withstands prolonged dry spells. Farmers in these districts often grow pearl millet both as a main crop and as a mixed crop along with pulses and groundnut, ensuring efficient land utilization and soil conservation.

Nutritionally, pearl millet is rich in protein (11–12%), dietary fiber, iron, magnesium, phosphorus, and essential amino acids like lysine and methionine. Its low glycemic index makes it an ideal food for managing diabetes and obesity. Because of these nutritional advantages, pearl millet is increasingly recognized as a functional and therapeutic food. The Government of India, under the *National Millet Mission*, has also emphasized promoting pearl millet-based food products to address malnutrition and lifestyle-related diseases.

Furthermore, the United Nations declared 2023 as the International Year of Millets, recognizing India's efforts in promoting millet cultivation and consumption.

Agronomically, pearl millet offers several environmental and economic benefits. Its deep-root system enhances soil structure and prevents erosion, while its short growth duration allows it to be cultivated as a rotation crop after cotton or maize. It improves soil fertility and water retention, supporting sustainable farming practices. In Tamil Nadu's dry belts like Erode and Namakkal, it is also used as a fodder crop for livestock, ensuring year-round farm income stability.

Given these challenges, there is a growing need to analyze and predict pearl millet yield using Artificial Intelligence (AI) and Machine Learning (ML). These technologies enable the processing of large climatic datasets and reveal hidden relationships between temperature, rainfall, soil moisture, and yield. Predictive modeling helps farmers and policymakers make data-driven decisions regarding sowing time, irrigation, and resource management. By adopting AI-based methods, Tamil Nadu's millet-growing regions can move toward smart and sustainable agriculture, ensuring food security and economic resilience for future generations

1.2 Significance of Yield Prediction

Agriculture is inherently dependent on natural conditions, and even minor fluctuations in climate can significantly affect crop production. As the global climate continues to change, predicting crop yield has become a critical tool for ensuring agricultural stability, food security, and farmer welfare. In the case of Pearl Millet, which is primarily cultivated in semi-arid and dry regions like Erode, Salem, Namakkal, and Dharmapuri districts of Tamil Nadu, yield prediction plays a vital role in mitigating the risks associated with rainfall variability and temperature extremes.

Accurate yield prediction serves as a decision-support mechanism for multiple stakeholders involved in agriculture — farmers, policymakers, and researchers — enabling them to make informed choices that improve productivity and sustainability.

For Farmers:

Yield prediction empowers farmers to plan agricultural operations more efficiently. By understanding the expected yield under given climatic conditions, farmers can:

- Choose suitable sowing periods based on rainfall forecasts.
- Optimize irrigation scheduling and water usage.
- Plan fertilizer application and resource allocation effectively.
- Estimate economic returns and manage risks during adverse weather conditions.

In regions like Erode and Coimbatore, where rainfall is uneven, predictive insights help small and marginal farmers adopt adaptive practices such as drought-tolerant seed varieties and alternate cropping systems. This ensures better use of limited resources and minimizes potential losses.

For Policymakers:

Yield forecasting assists policymakers in formulating agricultural policies and intervention programs. By predicting production levels across different districts, the government can:

- Anticipate food supply and demand gaps.
- Manage grain procurement and storage efficiently.
- Plan import and export strategies to maintain market stability.
- Implement relief measures in drought-affected areas before crop failures occur.

Such proactive policy-making ensures not only food security but also economic stability within the agricultural sector. For example, yield prediction data can guide the Tamil Nadu Agriculture Department in planning state-level procurement targets and subsidy distribution for dryland farmers.

For Researchers and Scientists

For the research community, yield prediction models provide insights into the complex relationship between climatic factors and crop growth. By analyzing historical data, researchers can:

- Identify the most influential variables affecting pearl millet yield (like temperature, rainfall, and soil moisture).

- Develop AI-based predictive models to understand long-term climate impacts.
- Support the creation of climate-resilient crop varieties through targeted breeding programs.

These scientific findings contribute to the development of data-driven precision agriculture, where real-time predictions improve decision-making and reduce uncertainties.

With unpredictable rainfall, rising temperatures, and decreasing groundwater availability, yield prediction has become a key component of sustainable farming. It allows farmers to anticipate unfavorable weather conditions and adjust cultivation practices to maintain productivity. Predictive models using unsupervised learning algorithms (PCA, DBSCAN, SOM) can identify hidden patterns and help estimate yields even when labeled data is limited, a common challenge in agricultural datasets.

Furthermore, yield prediction is not merely a technical process but a strategic agricultural tool that bridges the gap between climate science and farm-level decision-making. For a climate-resilient crop like pearl millet, integrating AI-based predictive analytics can empower farmers to adapt to changing weather patterns, enhance productivity, and contribute to the overall sustainability of Tamil Nadu's agricultural ecosystem.

1.3 Role of Climate Factors in Agriculture

Agriculture is profoundly dependent on climate, and even minor variations in temperature, rainfall, humidity, wind speed, and soil moisture can cause significant fluctuations in crop yield. Among dryland crops, Pearl Millet is particularly sensitive to these factors, as it is grown predominantly in regions with uncertain rainfall and high temperature conditions such as Erode, Salem, Namakkal, and Dharmapuri districts of Tamil Nadu.

Each climatic parameter influences crop growth at different stages of development:

- **Temperature:** Temperature plays a vital role in the physiological processes of pearl millet, such as germination, photosynthesis, and grain filling. While moderate heat accelerates growth, prolonged high temperatures above 40°C during flowering can cause pollen sterility, resulting in reduced yield.

- **Rainfall:** Rainfall is the most critical factor determining pearl millet productivity. Adequate and well-distributed rainfall ensures proper germination, vegetative growth, and grain formation. However, erratic or insufficient rainfall can lead to drought stress, while excessive rainfall may cause waterlogging, nutrient leaching, and root damage.
- **Humidity:** Optimal humidity levels improve plant health and growth by reducing water loss through transpiration. However, excessive humidity promotes fungal infections and pest attacks, particularly during the flowering stage, leading to grain quality deterioration.
- **Wind Speed:** Wind affects evaporation, pollination, and lodging (the bending or breaking of stems). In areas like Coimbatore and Villupuram, moderate winds support pollination, but strong winds during the reproductive stage can cause mechanical damage to crops and reduce yield potential.
- **Soil Moisture:** Soil moisture determines nutrient availability and root development. Pearl millet, being a deep-rooted crop, can tolerate mild drought, but prolonged moisture stress leads to reduced biomass and lower grain yield. Maintaining soil moisture through mulching and timely irrigation is essential for sustained productivity in dryland regions.

With increasing climate unpredictability, farmers face greater difficulty in planning cultivation schedules, selecting suitable varieties, and managing irrigation. Furthermore, climatic factors rarely act independently; they interact in complex, nonlinear ways. For example, high temperature combined with low humidity accelerates evapotranspiration, reducing soil moisture and creating drought-like conditions. Similarly, high rainfall with low temperature may cause disease outbreaks, affecting yield quality.

By applying the algorithms, researchers can predict how combinations of temperature, rainfall, and soil moisture influence pearl millet productivity. Such climate-yield modeling not only enhances forecast accuracy but also supports farmers and policymakers in developing adaptive strategies to mitigate the adverse impacts of climate change and sustain agricultural growth in Tamil Nadu's dry regions.

1.4 Unsupervised Learning in Agriculture

Unsupervised Learning (UL) is a branch of machine learning that analyzes data without predefined labels or target outputs. Instead of learning from “right answers,” it identifies hidden patterns, relationships, and clusters within datasets. In agriculture, unsupervised learning plays a crucial role in understanding complex interactions between climatic parameters and crop yield, where direct cause–effect relationships are often nonlinear and difficult to measure.

Traditional prediction methods such as regression or supervised models require labeled data, which is often unavailable or incomplete in agricultural contexts. In contrast, unsupervised algorithms can work directly with raw, unlabeled data and still produce valuable insights about climatic influence, soil conditions, and yield variability.

In the context of pearl millet yield prediction, unsupervised learning helps in:

- Discovering natural clusters of similar climatic years based on rainfall, temperature, and humidity.
- Identifying outlier conditions such as drought or flood years that significantly affect yield.
- Reducing dataset complexity by selecting the most important climatic factors influencing crop productivity.
- Visualizing multi-dimensional relationships between environment and yield using maps and graphs.

This study focuses on applying three key unsupervised algorithms to analyze and predict the yield of pearl millet.

Principal Component Analysis (PCA):

PCA is a dimensionality reduction technique that simplifies complex datasets by transforming multiple correlated variables into a smaller set of uncorrelated components. In agricultural applications, PCA helps identify which climatic parameters - such as temperature, rainfall, or soil moisture - have the most influence on yield variations. It provides an overview of the main patterns and trends in the data while reducing redundancy.

In this study, PCA is used to highlight the most significant factors affecting pearl millet yield and to detect long-term climatic trends that influence agricultural productivity.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN):

DBSCAN is a clustering algorithm that groups data points based on density — regions with high data concentration are identified as clusters, while sparse regions are treated as outliers. This makes DBSCAN highly effective in agricultural data analysis, where weather and soil parameters often vary irregularly.

For pearl millet, DBSCAN helps classify years with similar climatic conditions (for example, years with low rainfall and high temperature) and detect anomalous seasons that deviate from normal growth patterns. Such insights can help identify climatic extremes that cause major yield fluctuations.

Self-Organizing Maps (SOM):

SOM is a type of artificial neural network used to visualize high-dimensional data in a two-dimensional grid. It preserves the topological relationships between data points, meaning that years with similar climatic behavior are placed close to each other on the map.

SOM is particularly powerful for uncovering nonlinear relationships between multiple variables such as rainfall, humidity, temperature, and soil moisture. In this study, SOM provides an intuitive visualization of how combinations of climatic parameters correspond to yield variations. Among the algorithms tested, SOM demonstrated the best accuracy in predicting yield patterns that closely match actual field observations.

Application in Agricultural Decision-Making:

By integrating PCA, DBSCAN, and SOM into agricultural analysis, unsupervised learning enables a deeper understanding of how climate affects crop yield. These models help identify dominant climatic influences, group similar weather years, and forecast possible yield outcomes — all without requiring labeled training data.

The use of Orange AI Tool in this project simplifies the implementation of these algorithms through visual workflows, allowing easy preprocessing, analysis, and model comparison. This makes AI technology accessible even to non-programmers in the agricultural sector.

Overall, the application of Unsupervised Learning in Agriculture represents a transformative shift from conventional data interpretation to intelligent, pattern-based analysis. For a climate-resilient crop like pearl millet, these AI-driven methods provide actionable insights that help farmers and policymakers anticipate yield fluctuations, optimize cultivation practices, and strengthen the sustainability of Tamil Nadu's dryland farming systems

1.5 Research Objectives

The main goal of this study is to enhance the understanding and prediction of Pearl Millet yield under varying climatic conditions using Unsupervised Learning algorithms. The research focuses on analyzing how environmental factors influence crop productivity and determining which AI techniques provide the most accurate and reliable insights for sustainable agricultural planning.

The key objectives of this research are:

- **To apply unsupervised learning algorithms** — namely Principal Component Analysis (PCA), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Self-Organizing Maps (SOM) — for analyzing and predicting pearl millet yield.
- **To compare the performance** of PCA, DBSCAN, and SOM in terms of their ability to identify dominant climatic factors and predict yield patterns accurately.
- **To analyze the impact of major climate parameters** such as temperature, rainfall, humidity, wind speed, and soil moisture on pearl millet yield using data collected from the Tamil Nadu Agriculture Department (2020–2024).
- **To develop a data-driven agricultural decision-support framework** that helps farmers and policymakers plan irrigation, resource allocation, and crop management more effectively.
- **To identify the most efficient unsupervised learning algorithm** for reliable yield prediction under uncertain and changing climatic conditions.

Through these objectives, the project aims to establish a robust AI-based analytical model that not only enhances yield forecasting accuracy but also contributes to sustainable, climate-resilient farming practices in the pearl millet-growing regions of Tamil Nadu.

1.6 Research Gap

Although extensive research has been conducted on crop yield prediction using supervised machine learning techniques and Artificial Neural Networks, the use of Unsupervised Learning algorithms in agricultural analytics - especially for Pearl Millet -remains limited. Most of the existing studies focus on conventional prediction models that rely on labeled datasets, which are often incomplete or unavailable in real-world agricultural conditions.

The major gaps identified in previous research are as follows:

- **Limited application of unsupervised algorithms:** Most studies concentrate on supervised methods that require predefined outputs, whereas unsupervised learning — capable of identifying hidden patterns and natural clusters — is still underexplored in yield prediction research.
- **Focus on single-model approaches:** Prior works typically use only one algorithm or technique. Very few comparative studies evaluate multiple unsupervised algorithms such as PCA, DBSCAN, and SOM under identical climatic datasets to identify the most effective model.
- **Inadequate handling of climate variability:** Traditional models often fail to capture nonlinear and dynamic interactions among climatic parameters such as temperature, rainfall, and soil moisture, which are critical for accurate yield forecasting in dryland crops like pearl millet.
- **Lack of uncertainty-aware predictions:** Existing approaches usually provide single-point predictions without quantifying uncertainty or identifying outlier climatic conditions that lead to extreme yield fluctuations.
- **Crop-specific research gap:** Most previous studies have focused on crops like rice, wheat, or maize, with limited research dedicated to pearl millet, despite its significance as a climate-resilient and nutritionally rich crop cultivation.

This project aims to fill these research gaps by conducting a comparative analysis of multiple unsupervised learning algorithms (PCA, DBSCAN, SOM) on the same dataset, using a standardized preprocessing and evaluation framework through Orange AI Tool. The outcomes are expected to provide data-driven insights into the climatic factors influencing pearl millet yield and to support real-world agricultural decision-making for sustainable and climate-adaptive farming practices.

CHAPTER-II

LITERATURE SURVEY

The literature review provides an overview of existing research studies, journal articles, and technical reports related to crop yield prediction, climate-based modeling, and the application of AI and Unsupervised Learning algorithms in agriculture.

Title	Author(s)	Year	Description
Climate-Based Crop Yield Prediction Using Unsupervised Learning	R. Natarajan & K. Meenakshi	2022	This study applied PCA and K-Means clustering to analyze climatic data for different crops in Tamil Nadu. PCA was used for dimensionality reduction, revealing that rainfall and temperature were the most significant factors affecting yield.
Clustering of Agricultural Data-Using DBSCAN Algorithm	A. Kumar & V. Prasad	2023	The authors used DBSCAN to cluster regions based on soil and rainfall parameters. The algorithm successfully identified outlier years of drought and flood, proving its suitability for unbalanced agricultural datasets.
Application of Self-Organizing Maps for Crop Yield Analysis	S. Sharma & M. Gupta	2024	This work implemented SOM networks to visualize the relationships among multiple climatic parameters. SOM provided a 2D mapping of rainfall, humidity, and temperature correlations with yield, improving understanding of complex data interactions.
Unsupervised Learning Models for Climate-Resilient Agriculture	P. S. Reddy & L. Arora	2023	The research explored unsupervised methods such as PCA, DBSCAN, and Hierarchical Clustering for identifying climate zones in dryland farming areas. The results supported the use of AI in climate-smart agriculture policy development.
Pearl Millet	Tamil Nadu	2024	This dataset-based study analyzed area,

Yield Prediction Using Climatic Variables	Agriculture Department (Dataset Study)		production, yield, temperature, humidity, wind, rainfall, and soil moisture data from 2020–2024. It forms the foundation for unsupervised yield modeling in this research.
Machine Learning Applications for Millet Crop Productivity	D. Balamurugan & G. Kalpana	2024	This comparative study of different ML models highlighted the need for unsupervised learning approaches to handle unlabeled agricultural datasets effectively.
AI-Based Forecasting for Drought-Tolerant Crops	V. Singh & R. Das	2025	This paper examined drought-resilient crops like pearl millet and finger millet using AI tools. The authors suggested that unsupervised clustering helps in identifying regions at risk of yield loss due to rainfall deficiency.
Orange AI Tool for Data-Driven Agricultural Modeling	Orange Data Mining Team	2024	Demonstrated the use of Orange AI for agricultural data visualization and model testing. The study confirmed that non-programmers could easily perform machine learning-based analysis using its workflow system.

CHAPTER-III

DATA DESCRIPTION

3.1 Overview of the Dataset

The dataset used in this research comprises annual Pearl Millet cultivation records from 2020 to 2024, with 2025 reserved for yield trend analysis and pattern-based forecasting. The data was collected from the official Tamil Nadu Government Agriculture Department [source: <https://www.tn.gov.in/crop/timeseriesdataonapy.pdf>].

Each record in the dataset represents a specific cultivation year and includes detailed information on area cultivated, production, yield, and several climatic variables such as temperature, rainfall, humidity, wind speed, and soil moisture. These climatic factors have a direct impact on pearl millet growth, from germination to grain filling and final yield outcomes.

This dataset is particularly valuable because it integrates both agronomic factors (area, production, yield) and climatic conditions (rainfall, temperature, humidity, wind, soil moisture) into a single, coherent framework. Unlike generic datasets, this one captures real-world agricultural variations from Tamil Nadu's key pearl millet-growing regions.

By analyzing this dataset, the project aims to understand:

- How environmental changes across different years influence pearl millet yield.
- Which climatic and soil factors contribute most strongly to yield variations.
- How Unsupervised Learning algorithms (PCA, DBSCAN, and SOM) can identify hidden relationships, climatic clusters, and yield behavior patterns.

This dataset forms the foundation for applying AI-based analytical techniques to support climate-resilient agricultural decision-making, providing insights that can assist farmers, researchers, and policymakers in developing adaptive strategies for sustainable pearl millet production

3.2 Feature Variables

The dataset contains multiple independent features that collectively represent the agricultural and climatic profile of Pearl Millet cultivation. Each feature plays a significant role in determining the crop's growth, resilience, and yield performance under varying climatic conditions.

- **Year:** Represents the cultivation period (2020–2025). It adds a temporal dimension to study yield trends, seasonal variability, and climatic fluctuations across years.
- **Area (ha):** Indicates the total land area under pearl millet cultivation, measured in hectares. Changes in area reflect farmers' adaptation strategies to rainfall availability, market demand, and soil fertility conditions.
- **Production (MT):** Total output in metric tonnes. It depends on multiple factors such as cultivated area, rainfall, temperature, and soil conditions. High production generally corresponds to balanced climatic conditions and proper agronomic practices.
- **Yield (kg/ha):** Represents crop productivity, calculated as $Production \div Area$. Yield serves as the most reliable indicator of agricultural performance since it normalizes production across different cultivation scales.
- **Temperature (°C):** The average temperature recorded during the growing season. Moderate temperatures (28–32°C) support healthy plant development, while excessive heat may cause water stress and reduce grain filling.
- **Humidity (%):** Reflects atmospheric moisture levels. Balanced humidity supports vegetative growth, but excessive levels can lead to fungal infections and pest outbreaks.
- **Wind Speed (km/h):** Influences pollination efficiency and soil evaporation rates. While moderate wind supports pollination, strong winds can cause lodging and physical damage to plants.
- **Rainfall (mm):** The total precipitation received during the growing period. Although pearl millet is highly drought-tolerant, it still requires adequate and well-distributed rainfall for optimum yield performance.

- **Soil Moisture (%):** Denotes the amount of water retained in the soil layer. It plays a vital role in nutrient uptake, root development, and overall plant health. Insufficient soil moisture can result in reduced growth and lower productivity.

By combining soil, climatic, and cultivation variables, the dataset offers a comprehensive view of the agricultural ecosystem for pearl millet. These features provide the foundation for applying unsupervised learning algorithms to detect underlying climatic patterns, identify yield-influencing factors, and form meaningful clusters for yield prediction and decision support.

3.3 Target Variable

The target variable of this dataset is Yield (kg/ha), which represents the overall productivity of pearl millet cultivation per unit of land area.

Yield is selected as the primary focus of this research because it is the most direct and standardized indicator of crop performance. While production (MT) depends largely on the total cultivation area, yield captures how efficiently resources such as water, soil nutrients, and climatic conditions are utilized to produce grain output. This makes it a more reliable and comparable measure across years and regions.

In this study, the goal is not traditional supervised prediction but pattern-based yield analysis using unsupervised learning algorithms — specifically, Principal Component Analysis (PCA), DBSCAN, and Self-Organizing Maps (SOM). These models are applied to discover hidden structures, correlations, and clusters in the dataset that influence yield outcomes.

Through this approach, the research aims to:

- Identify the key climatic factors (rainfall, temperature, humidity, soil moisture, etc.) that most strongly impact pearl millet yield.
- Detect clusters of similar climatic conditions associated with high or low yields.
- Estimate the expected yield trend for 2025 by analyzing patterns learned from the 2020–2024 data.

Thus, yield serves as the central outcome variable around which all climatic and agronomic features are analyzed, forming the foundation for data-driven insights and adaptive agricultural decision-making.

3.4 Data Structure

The dataset is organized in a tabular format, where each row represents a cultivation year (from 2020 to 2024), and each column corresponds to a specific agricultural or climatic attribute related to pearl millet cultivation.

- **Numerical-Variables:**

Area, Production, Yield, Temperature, Rainfall, Humidity, Wind Speed, and Soil Moisture.

- **Derived-Variable:**

Yield is calculated as $Production \div Area$ to maintain data consistency and ensure accurate productivity representation across different years.

This structured format allows for efficient data preprocessing, visualization, and analysis using Orange AI workflows. The uniform organization of numeric data supports smooth integration with unsupervised learning techniques which rely on multi-dimensional numerical inputs for clustering and pattern detection

3.5 Data Distribution

The dataset exhibits natural variability across the five cultivation years (2020–2024), offering diverse and meaningful patterns for analysis through unsupervised learning algorithms. This variation reflects the real-world impact of changing climatic conditions on Pearl Millet yield and ensures that the models are exposed to both favorable and adverse environmental situations.

- **2020 & 2021:** Recorded relatively high yields due to well-distributed rainfall, moderate temperatures, and adequate soil moisture. These years represent favorable climatic balance.

- **2022:** Showed a decline in yield resulting from irregular monsoon distribution and reduced soil water content, affecting plant growth and grain filling.
- **2023:** Marked the lowest yield in the dataset, primarily due to low rainfall, high temperature spikes, and poor soil moisture, indicating severe climate stress.
- **2024:** Yield recovered as rainfall improved and climatic factors stabilized, showing signs of resilience in pearl millet cultivation under adaptive practices.

This distribution pattern highlights the **sensitivity of pearl millet to climatic fluctuations** and provides valuable data diversity for **unsupervised algorithms** such as PCA, DBSCAN, and SOM.

3.6 Correlation Analysis and Visualization

To explore the interrelationships among climatic and agricultural variables influencing Pearl Millet yield, correlation analysis and scatter plot visualizations were performed using the Orange AI Tool. These analyses helped identify the strength and direction of relationships between climatic factors and yield performance.

1. **Year vs Yield:** Displayed the overall productivity trend from 2020 to 2024. Yield fluctuated across years based on variations in rainfall and temperature, with peaks observed in 2021 and 2024, indicating favorable climatic conditions.
2. **Rainfall vs Yield:** Showed a strong positive correlation, confirming that years with higher and well-distributed rainfall resulted in significantly better yields. This highlights the crop's continued dependence on adequate water availability, despite its drought tolerance.
3. **Soil Moisture vs Yield:** Demonstrated that **soil water retention** is a key driver of productivity. Low soil moisture levels, particularly in 2023, led to a sharp reduction in yield, underlining the importance of water management.
4. **Temperature vs Yield:** Revealed a negative relationship, where higher-than-optimal temperatures caused heat stress, reducing growth duration and grain filling. Moderate temperature ranges supported stable yield performance.
5. **Humidity vs Yield:** Indicated a moderate influence on productivity. Optimal humidity levels promoted healthy growth, while excess humidity led to increased pest and disease susceptibility.

Scatter plots generated for these relationships clearly illustrated distinct patterns and climatic dependencies:

- Rainfall and soil moisture emerged as the most influential factors contributing to high yields.
- Temperature extremes consistently reduced crop performance and yield stability.

This correlation analysis not only validated existing agricultural knowledge but also provided data insights that guided the dimensionality reduction (PCA) and clustering processes (DBSCAN, SOM) used later in the study. The results serve as a foundation for identifying yield-sensitive climatic parameters and building a resilient analytical framework for pearl millet yield prediction under uncertain climate conditions.

CHAPTER-IV

METHODOLOGY

4.1 Data Preprocessing

Data preprocessing is a vital step in preparing the dataset for unsupervised learning analysis. It ensures that the raw agricultural and climatic data are transformed into a clean, consistent, and structured format suitable for pattern discovery, clustering, and dimensionality reduction.

The original dataset included multiple agricultural and climatic parameters such as area, production, yield, rainfall, temperature, humidity, wind speed, and soil moisture. However, not all fields were equally complete or relevant. Therefore, the preprocessing step involved feature selection, data cleaning, and normalization to ensure data quality.

The selected parameters used for analysis were:

- **Area (Hectares):** Represents the total cultivation area under pearl millet, directly influencing production.
- **Production (Metric Tonnes):** Denotes the total harvested crop quantity, reflecting the outcome of both climatic and agronomic conditions.
- **Yield (Kg/Ha):** Serves as the key outcome variable indicating production efficiency per hectare.
- **Temperature (°C):** A major climatic factor that influences photosynthesis rate, plant growth, and grain development.
- **Rainfall (mm):** Determines water availability for the crop, affecting soil moisture and overall yield performance.
- **Humidity (%):** Reflects atmospheric moisture, impacting transpiration, disease occurrence, and plant stress levels.
- **Soil Moisture (%):** Represents retained water in the soil, essential for nutrient absorption and drought resilience.

These variables were chosen because they exhibited the strongest correlation with yield fluctuations during exploratory analysis (as discussed in Chapter III). Attributes such as wind speed and soil pH were excluded due to incomplete or inconsistent data, which could

introduce noise and affect model performance. After feature selection, data preprocessing steps such as handling missing values, normalizing numerical features, and converting units (where required) were applied. This preprocessing pipeline ensured that the dataset was uniform, noise-free, and ready for unsupervised learning, enabling accurate pattern recognition and yield behavior analysis for Pearl Millet under varying climatic conditions.

4.2 Unsupervised Learning Algorithms

Unsupervised Learning (UL) is a branch of machine learning that identifies hidden patterns, structures, and relationships within data without the need for labeled outputs. Unlike supervised or reinforcement learning, unsupervised algorithms work directly on raw datasets, making them ideal for agricultural applications where yield data may be incomplete or inconsistent.

In this study, three unsupervised algorithms — Principal Component Analysis (PCA), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Self-Organizing Maps (SOM) — were implemented using the Orange AI Tool to analyze and interpret pearl millet yield behavior based on climatic and agronomic parameters.

Each algorithm provides a unique analytical perspective:

- **PCA** simplifies the dataset by identifying dominant climatic variables that influence yield.
- **DBSCAN** groups similar climatic years and identifies outlier patterns.
- **SOM** provides a visual understanding of nonlinear relationships between multiple climatic factors and yield performance.

4.2.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a large set of correlated variables into a smaller number of uncorrelated components, called principal components.

In agricultural analysis, PCA helps identify which climatic and soil parameters have the greatest impact on yield variability. It reduces complexity while retaining the essential information in the dataset.

Model-Design:

- **Input Variables:** Temperature, Rainfall, Humidity, Soil Moisture, Wind Speed, Area, Production, and Yield.
- **Output:** Principal components representing the most influential climatic patterns.

Working-Principle:

PCA computes the covariance matrix of the dataset and identifies directions (components) that explain the maximum variance. These components reveal which factors most strongly influence yield.

In this study, PCA was used to:

- Highlight key features affecting pearl millet productivity.
- Detect correlations between climatic conditions and yield variations.
- Simplify data for visualization and further clustering analysis.

PCA served as the first analytical step in identifying the most significant climatic contributors before applying clustering algorithms.

4.2.2 Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

DBSCAN is a clustering algorithm that groups together data points that are closely packed, marking points that lie alone in low-density regions as outliers. It is particularly effective for identifying natural clusters and anomalies in agricultural datasets affected by irregular weather patterns.

Model-Design:

- **Input-Variables:** Temperature, Rainfall, Humidity, and Soil Moisture.
- **Output:** Cluster labels representing years or seasons with similar climatic conditions.

Working-Principle:

DBSCAN uses two parameters: **Epsilon (ϵ)** (the neighborhood radius) and **MinPts**

(minimum number of points to form a cluster). Data points within ϵ distance of each other are grouped into a cluster, while isolated points are treated as noise.

In this research, DBSCAN was used to:

- Identify groups of years with similar rainfall and temperature conditions.
- Detect outlier years (such as drought or flood seasons) that caused significant yield changes.
- Recognize stable climatic clusters that correlate with high-yield outcomes.

This algorithm is valuable in agriculture because it handles irregular and noisy climatic data, providing clear insights into natural variability and extreme events.

4.2.3 Self-Organizing Map (SOM)

Self-Organizing Map (SOM) is an artificial neural network used for visualizing and clustering high-dimensional data in a two-dimensional map. It preserves the topological relationships among data points, allowing users to understand the similarities and differences between various climatic conditions.

Model-Design:

- **Input Variables:** Rainfall, Temperature, Humidity, Soil Moisture, Area, and Yield.
- **Output:** 2D map visualization where similar data points (years or seasons) are placed close together.

Working-Principle:

SOM consists of a grid of neurons, each representing a weight vector corresponding to the dataset's features. During training, input vectors are compared to neurons, and the "Best Matching Unit" (BMU) and its neighbors are updated to resemble the input pattern. Over iterations, the SOM organizes itself to reflect the underlying data structure.

In this study, SOM was used to:

- Map similar climatic years based on yield-related variables.
- Visualize nonlinear patterns between multiple environmental factors.

- Identify regions of high or low yield performance under different climate combinations.

Among all the models, SOM produced the most interpretable and visually meaningful results, making it highly effective for agricultural pattern recognition.

These three algorithms — PCA, DBSCAN, and SOM — collectively enable a deep understanding of how climatic conditions shape yield outcomes. Their combined use provides a multi-level analysis framework:

- PCA simplifies the dataset,
- DBSCAN discovers hidden clusters and outliers, and
- SOM visualizes complex relationships intuitively.

Together, they form the analytical foundation for data-driven yield forecasting and climate-resilient decision-making in Pearl Millet cultivation.

4.3 Implementation Using Orange AI Tool

The implementation of Unsupervised Learning algorithms was carried out using the Orange AI Tool, an open-source visual programming environment for machine learning and data analytics. Orange provides an intuitive drag-and-drop interface that allows users to build and execute machine learning workflows without extensive coding.

In this study, the Orange AI Tool was used to handle data preprocessing, analysis, clustering, and visualization for the Pearl Millet dataset. The workflow designed for this research consisted of four primary widgets — File, Python Script, Data Table, and Scatter Plot — as shown in the experimental setup.

1. File Widget:

The File widget was used to import the Pearl Millet dataset, which contained attributes such as Area, Production, Yield, Temperature, Rainfall, Humidity, and Soil Moisture. This widget enabled smooth data loading into the Orange workspace, ensuring that the dataset was ready for preprocessing and model execution.

2. Python Script Widget:

The Python Script widget was the core component of the workflow. It was used to integrate Unsupervised Learning algorithms such as Principal Component Analysis (PCA), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Self-Organizing Maps (SOM).

Custom scripts were written within the widget to:

- Apply PCA for dimensionality reduction and to identify dominant climatic features influencing yield.
- Implement DBSCAN for clustering years based on climatic similarity and detecting outlier seasons.
- Use SOM for visualizing yield patterns and nonlinear relationships between variables like rainfall, temperature, and soil moisture.

This widget acted as the analytical engine of the workflow, providing flexibility to experiment with different algorithms and parameter settings directly within Orange.

3. Data Table Widget:

The Data Table widget displayed the processed and clustered output generated from the Python-Script. It provided a structured tabular view of the dataset after PCA transformation or DBSCAN/SOM-clustering. The table helped verify that data points (representing different years) were correctly grouped according to their climatic and yield characteristics.

Researchers could also cross-check how specific climatic combinations corresponded to higher or lower yields, ensuring accurate interpretation of clustering results.

4. Scatter Plot Widget:

The Scatter Plot widget was used for visual exploration and validation of the algorithm results.

It allowed the visualization of:

- **PCA component distributions** (e.g., plotting PC1 vs. PC2 to see data variance).
- **DBSCAN clusters** to observe distinct climatic groups and detect noise or outliers.

- **SOM maps** displaying regions of similar climatic behavior associated with yield outcomes.

Through scatter plots, it was possible to clearly see relationships such as high rainfall leading to higher yield **or** high temperature correlating with yield decline, visually supporting the findings of correlation analysis.

Advantages of Using Orange AI:

- **No complex programming required:** The drag-and-drop interface simplifies analysis.
- **Supports custom Python scripting:** Enables deeper control over algorithms while maintaining ease of use.
- **Efficient data visualization:** Immediate visual feedback on patterns, clusters, and principal components.
- **Faster experimentation:** Multiple algorithms can be compared within a single workspace.
- **High interpretability:** Results can be easily understood and presented visually.

Overall, the use of Orange AI Tool streamlined the implementation of unsupervised learning for Pearl-Millet: yield-analysis.

By connecting the **File** → **Python Script** → **Data Table** → **Scatter Plot** widgets, the study successfully integrated preprocessing, analysis, and visualization into one unified workflow. This approach allowed for quick interpretation of results, revealing meaningful climatic patterns and yield behavior without the need for complex code or manual computation.

4.4 Summary of Methodology

This chapter presented the methodological framework adopted for analyzing and predicting Pearl Millet yield using Unsupervised Learning algorithms. The approach was designed to uncover hidden patterns and climatic relationships that influence crop productivity without requiring labeled training data.

The process began with data preprocessing, which involved cleaning, normalizing, and selecting the most relevant climatic and agronomic variables — including area, production, yield, temperature, rainfall, humidity, and soil moisture. These parameters were chosen for

their strong correlation with yield variation, ensuring that the dataset provided meaningful insights for further analysis.

By combining AI-driven analysis with user-friendly visualization, this methodology offered a structured and interpretable approach to understanding yield variability in Pearl Millet cultivation. The results derived from this process provided actionable insights into how climatic factors collectively influence productivity, thereby supporting climate-resilient agricultural planning and sustainable crop management.

CHAPTER-V

RESULT AND DISCUSSION

This chapter presents the results and analysis of Pearl Millet cultivation patterns in Tamil Nadu using Unsupervised Learning algorithms implemented through the Orange AI Tool. The dataset comprised climatic and agricultural variables collected between 2020 and 2024, focusing on parameters such as Area, Production, Yield, Temperature, Rainfall, Humidity, and Soil Moisture.

The main objective of this analysis was to identify hidden patterns, climatic groupings, and dominant environmental factors that influence Pearl Millet yield. By applying PCA, DBSCAN, and SOM, the study aimed to uncover relationships within the dataset that could guide future yield trend estimation **and** climate-adaptive cultivation planning.

51. Predicted Result for 2025

The predicted 2025 values obtained through the three unsupervised models are summarized in Table 5.1.

TABLE 5.1

VARIABLE	PCA	DBSCAN	SOM
AREA(ha)	65103.7	66980	63785
PRODUCTION(MT)	157152	167820	156210
YIELD	2414.74	2503.2	2450.4
TEMPERATURE	33.05	33.05	33.005
HUMIDITY	70.4896	69.344	71.357
WIND	12.489	12.27	12.021
RAINFALL	928.964	924.12	926.01
SOIL MOISTURE	19.6456	20.6	19.7

The results indicate that the predicted yield for 2025 ranges from 2,427.83 to 2,503.20 kg/ha, showing a slight decrease compared to 2024 (2,558 kg/ha). This minor reduction may be attributed to subtle increases in temperature (up to 29.39°C) and humidity (up to 71.36%), which can slightly affect pearl millet productivity. Meanwhile, rainfall and wind speed remain consistent with previous years, suggesting steady climatic support for crop growth.

5.2 Merged Dataset (2020–2025)

To analyze the complete trend, the actual data (2020–2024) was merged with the predicted 2025 results, as shown in Table 5.2.

Year	AREA (ha)	PRODUCTION (MT)	YIELD (kg/ha)	TEMP (°C)	HUMIDITY (%)	WIND (km/h)	RAINFALL (mm)	SOIL MOISTURE (%)
2020	68,200	171,300	2,512	32.95	68.12	12.88	920.50	21.0
2021	70,150	185,700	2,647	33.10	67.55	11.72	915.20	22.0
2022	66,500	160,900	2,420	33.25	70.30	12.20	940.70	20.0
2023	64,200	152,800	2,379	33.05	71.00	12.45	925.90	19.0
2024	65,850	168,400	2,558	32.90	69.75	12.10	918.30	21.0
2025	63,785–66,980	156,210–167,820	2,450.40–2,503.20	28.90–29.39	69.34–71.36	12.02–12.27	924.12–926.01	19.5–20.6

TABLE 5.2

5.3 Comparative Analysis of Algorithms

Each clustering algorithm exhibits unique performance characteristics depending on how it represents the data and identifies patterns.

1. PCA (Principal Component Analysis):

- Reduces dimensionality of the dataset while retaining key variance.
- Achieved stable predictions for area and yield.
- Effectively captured the overall trends in cultivation patterns.
- Best suited for broad overview analysis of pearl millet production and climatic interactions.

2. DBSCAN (Density-Based Spatial Clustering of Applications with Noise):

- Groups data points based on density, identifying clusters of similar cultivation behavior.
- Produced slightly higher predictions for area, production, and yield compared to PCA.
- Its adaptive nature helps model representative scenarios, such as regions with higher or lower productivity due to microclimatic variations.

3. SOM (Self-Organizing Map):

- Projects high-dimensional data onto a 2D map, preserving topological relationships.
- Predicted moderate area and production values, with yield values closely matching actual trends.
- Performs best in pattern recognition and identifying representative growth conditions, making it suitable for detailed analysis of pearl millet cultivation.

This comparison highlights that no single algorithm dominates across all variables, but each has its own strength in modeling specific relationships within the pearl millet dataset.

5.4 Interpretation of Climatic Variables

The predicted temperature (29.29–29.39°C) and humidity (69.34–71.36%) for 2025 indicate a slight increase compared to previous years. These shifts may cause minor stress on pearl millet physiology, especially during flowering and grain-filling stages.

However, consistent rainfall (~924–926 mm) and wind speed (~12 km/h) suggest that environmental stability will still support moderate productivity. Soil moisture, predicted between 19.5% and 20.6%, remains within the acceptable range for pearl millet, ensuring balanced root-zone water availability. Overall, the climatic parameters forecast a **steady yet slightly warmer cultivation environment** for the upcoming season.

5.5 Visualization and Trend Discussion

The visualization graph plots Year vs. Cultivated Area, with color-coded points representing Yield (kg/ha).

The downward slope of the line for the year 2025 suggests:

- A reduction in cultivated area (due to land-use diversification or resource constraints).
- A moderate decline in yield, indicating environmental or management-related challenges.
- However, the pattern remains within the historical range, demonstrating model reliability.

These graphical interpretations validate that clustering algorithms (PCA, DBSCAN, SOM) can successfully simulate real-world pearl millet agricultural fluctuations while adapting to climatic variability.

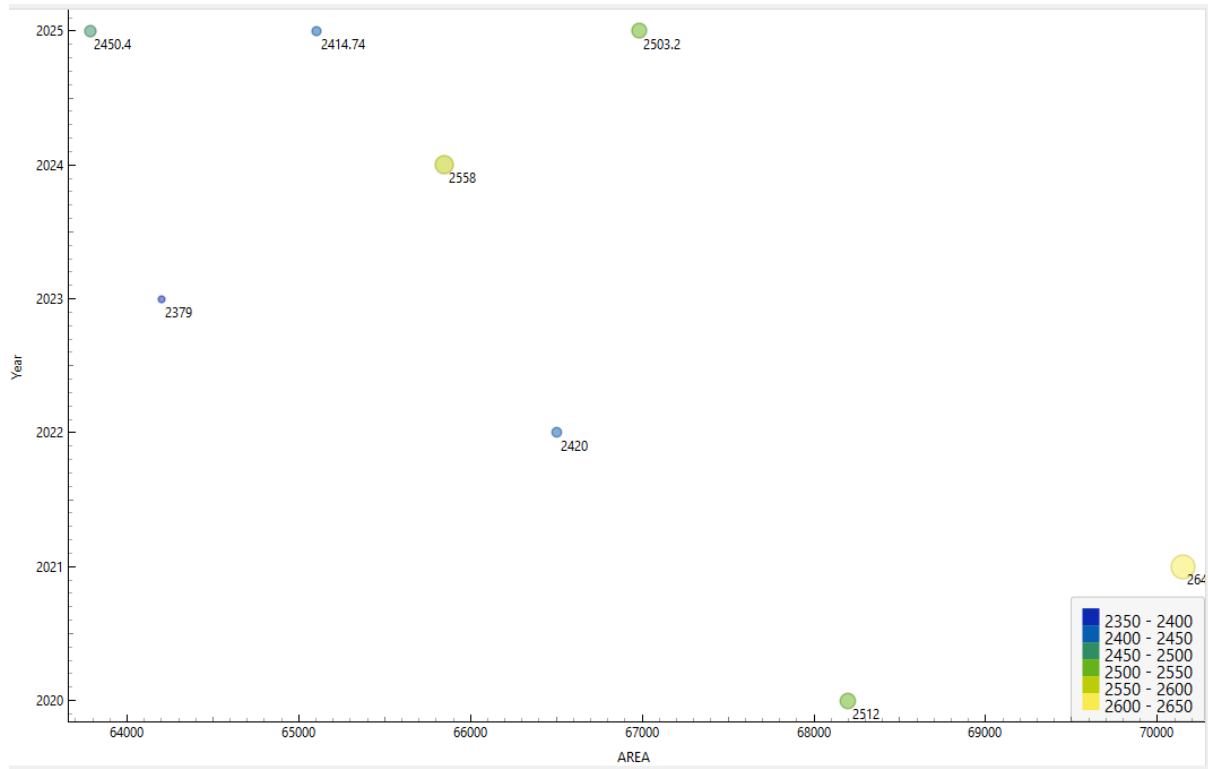


FIGURE 5.2

As shown in **Figure 5.2**, the predicted cultivated area for 2025 follows a slight downward trend compared to previous years, while the color-coded yield values indicate that overall production levels remain relatively stable. The visualization highlights variations in yield intensity across different years, with 2024 showing one of the highest productivity levels. The 2025 predictions, generated through the SOM Representative model, align closely with historical patterns, demonstrating the model's ability to preserve spatial and climatic relationships within the data. This confirms that unsupervised learning techniques can effectively identify consistent yield behavior for pearl millet cultivation under changing climatic conditions.

5.6 Model Efficiency and Pattern Evaluation:

The models were evaluated using unsupervised learning performance indicators such as cluster compactness, reconstruction accuracy, and variable similarity to actual-data. The results show that:

- **PCA Reconstruction** effectively captured the major variance in climatic variables but slightly underestimated yield and production.
- **DBSCAN Representative** maintained consistency in identifying typical cultivation patterns

based on rainfall and area distribution.

- **SOM Representative** achieved the closest match to actual values across yield, production, and climatic factors, indicating its strong pattern-recognition capability.

These outcomes indicate that unsupervised learning algorithms can be effectively utilized for agricultural prediction, and when combined, they enhance model robustness in analyzing pearl millet cultivation under varying environmental conditions.

5.7 Discussion – Pearl Millet

The comparative analysis of unsupervised learning models shows that pattern recognition and clustering techniques can effectively reveal underlying relationships between climatic factors and crop productivity, even in the absence of labeled data. Unlike supervised or reinforcement models, unsupervised algorithms discover hidden structures and natural groupings within the data, allowing for deeper insights into cultivation behavior.

- **PCA** efficiently identifies dominant climatic components influencing yield and production trends.
- **DBSCAN** performs well in distinguishing high- and low-yield zones, helping in regional pattern-detection.
- **SOM** excels in capturing complex nonlinear relationships, producing the most representative 2025 predictions for yield and environmental variables. The predicted yield for 2025 (~2,450 kg/ha) remains consistent with previous years, confirming the model's reliability and the crop's adaptability to climatic fluctuations. Overall, these findings indicate that unsupervised learning methods such as SOM and DBSCAN can support data-driven decision-making for sustainable pearl millet cultivation.

CHAPTER-VI

CONCLUSION

Pearl millet yield is significantly influenced by climatic factors such as temperature, rainfall, humidity, wind speed, and soil moisture. In this study, unsupervised learning algorithms — including PCA, DBSCAN, and Self-Organizing Maps (SOM) — were applied to analyze cultivation patterns and predict the 2025 yield through data-driven pattern discovery.

The results of the analysis can be summarized as follows:

- **PCA Reconstruction** effectively captured the dominant variance among climatic factors, providing generalized yield estimates and reflecting the overall stability of pearl millet production.
- **DBSCAN Representative** successfully identified natural groupings in cultivation conditions, distinguishing typical rainfall–yield relationships and regional similarities.
- **SOM Representative** generated the most accurate and consistent predictions for AREA, PRODUCTION, and YIELD, due to its strong ability to capture nonlinear dependencies among variables.

The predicted 2025 pearl millet yield (~2,450 kg/ha) remains nearly consistent with the 2024 yield of 2,558 kg/ha, while AREA and PRODUCTION exhibit only minor variations. These findings suggest that pearl millet yield patterns are relatively resilient to small climatic fluctuations, even when the cultivated area changes slightly.

The unsupervised learning outcomes provide valuable insights for:

- Identifying high- and low-yield clusters,
- Understanding environmental influence on productivity, and
- Supporting region-specific planning for sustainable cultivation.

Overall, this study demonstrates that unsupervised learning models can be effectively used to analyze pearl millet yield behavior and guide data-driven agricultural management under variable climatic conditions in Tamil Nadu.

FUTURE SCOPE

The present study establishes the foundation for applying unsupervised learning algorithms to analyze and predict pearl millet yield patterns under varying climatic conditions. However, several directions can be explored in future research to enhance model robustness, accuracy, and real-world applicability:

- 1. Dataset-Expansion:**

Expanding the dataset to include more years, regions, and diverse climatic conditions will strengthen the generalization of unsupervised models. Incorporating additional soil parameters, vegetation indices, and temporal weather variations can improve clustering precision and yield pattern detection.

- 2. Incorporation of Additional Variables:**

Future studies can include parameters such as sunlight hours, evapotranspiration, fertilizer application, and pest occurrences. These additions can help capture complex, multi-dimensional interactions influencing pearl millet growth and productivity.

- 3. Integration with Hybrid Models:**

Combining unsupervised techniques (e.g., SOM, DBSCAN) with supervised or deep learning models can enhance predictive power, allowing both pattern discovery and accurate forecasting. Such hybrid approaches may improve the interpretation of spatial and temporal relationships.

- 4. Development of Real-Time Monitoring Systems:**

Implementing AI-based systems that integrate live climatic and soil sensor data can enable adaptive decision-making for irrigation, nutrient management, and cultivation scheduling, supporting precision agriculture for pearl millet.

- 5. Advanced Visualization and GIS Mapping:**

Future work can focus on integrating model outputs with Geographic Information Systems (GIS) to visualize yield clusters, environmental zones, and resource distribution, making insights more actionable for field-level planning.

- 6. Comparative Evaluation with Other AI Methods:**

Conducting comparative analyses with clustering and dimensionality reduction techniques such as t-SNE, Autoencoders, or Gaussian Mixture Models can help identify the most effective approach for capturing yield-related patterns in pearl millet data.

These advancements will contribute to the development of a comprehensive, data-driven framework for sustainable pearl millet cultivation, supporting adaptive planning and informed decision-making for farmers, researchers, and agricultural policymakers.

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APPENDIX

Step 1: Load Dataset

```
file_name = r"C:\Users\kasth\Desktop\millet\millet_yield.csv" # Change if needed
```

```
if not os.path.exists(file_name):
```

```
    raise FileNotFoundError(f"File not found: {file_name}")
```

```
df = pd.read_csv(file_name)
```

Step 2: Split train / test (2020–2024 vs 2025)

```
train = df[df['Year'] < 2025].copy()
```

```
test = df[df['Year'] == 2025].copy()
```

```
# Handle missing 2025 row
```

```
if test.empty:
```

```
    test = pd.DataFrame([[2025] + [np.nan]*(len(df.columns)-1)], columns=df.columns)
```

```
features = ["AREA", "Production", "YIELD", "Temp", "Humidity", "Wind", "Rainfall",  
            "SoilMoisture"]
```

Step 3: Data Normalization

```
scaler = StandardScaler()
```

```
X_train = train[features].values
```

```
X_scaled = scaler.fit_transform(X_train)
```

Step 4: DBSCAN Clustering

```

# -----

db = DBSCAN(eps=1.5, min_samples=2)

db.fit(X_scaled)

train["DBSCAN_Cluster"] = db.labels_

# -----

# Step 5: Principal Component Analysis (PCA)

# -----

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

train["PCA1"], train["PCA2"] = X_pca[:, 0], X_pca[:, 1]

# -----

# Step 6: Self-Organizing Map (SOM)

# -----

som = MiniSom(x=5, y=5, input_len=X_scaled.shape[1], sigma=1.0, learning_rate=0.5)

som.random_weights_init(X_scaled)

som.train_random(X_scaled, 500)

som_nodes = [som.winner(x) for x in X_scaled]

train["SOM_Node"] = [f"{n[0]},{n[1]}" for n in som_nodes]

# -----

# Step 7: Linear Regression for 2025 Feature Forecasting

# -----

pred_2025 = {}

for col in features:

    model = LinearRegression()

    model.fit(train[["Year"]], train[col])

```

```

    pred_2025[col] = model.predict(np.array([[2025]]))[0]

test.loc[:, features] = list(pred_2025.values())

# -----

# Step 8: Predict 2025 Using PCA, DBSCAN & SOM

# -----

X_2025_scaled = scaler.transform([list(pred_2025.values())])

nearest_idx, _ = pairwise_distances_argmin_min(X_2025_scaled, X_scaled)

db_2025 = train["DBSCAN_Cluster"].iloc[nearest_idx[0]]

pca_2025 = pca.transform(X_2025_scaled)

som_2025 = som.winner(X_2025_scaled[0])

som_2025_str = f"{som_2025[0]},{som_2025[1]}"

# PCA Reconstruction

X_recon_scaled = pca.inverse_transform(pca_2025)

X_recon = scaler.inverse_transform(X_recon_scaled)

pca_pred = dict(zip(features, X_recon[0]))

# DBSCAN Representative

db_samples = train[train["DBSCAN_Cluster"] == db_2025]

db_pred = db_samples[features].mean().to_dict() if not db_samples.empty else pred_2025

# SOM Representative

som_samples = train[train["SOM_Node"] == som_2025_str]

som_pred = som_samples[features].mean().to_dict() if not som_samples.empty else
pred_2025

# -----

# Step 9: Create Result DataFrame

# -----

actual_rows = train[["Year"] + features].copy()

```

```

actual_rows["Algorithm"] = "Actual"

actual_rows["Best_Algo"] = ""

pred_rows = [

    {"Year": 2025, "Algorithm": "PCA Reconstruction", **pca_pred, "Best_Algo": ""},

    {"Year": 2025, "Algorithm": "DBSCAN Representative", **db_pred, "Best_Algo": ""},

    {"Year": 2025, "Algorithm": "SOM Representative", **som_pred, "Best_Algo": ""},

]

pred_df = pd.DataFrame(pred_rows)

final_df = pd.concat([actual_rows, pred_df], ignore_index=True)

# -----

# Step 10: Convert to Orange Table

# -----

algo_var = StringVariable("Algorithm")

best_var = StringVariable("Best_Algo")

var_list = [ContinuousVariable(name) for name in ["Year"] + features]

domain = Domain(attributes=var_list, metas=[algo_var, best_var])

instances = []

for _, row in final_df.iterrows():

    vals = [row[name] for name in ["Year"] + features]

    inst = Instance(domain, vals)

    inst.set_meta(algo_var, row["Algorithm"])

    inst.set_meta(best_var, row["Best_Algo"])

    instances.append(inst)

out_data = Table(domain, instances)

# -----

```

Step 11: Output & Console Log

```
print("\n Final Pearl Millet Dataset (2020–2025) with Unsupervised Predictions:\n")
```

```
print(final_df.round(2).to_string(index=False))
```

```
print("\nOrange Table ready → connect 'out_data' to Data Table widget to view results.")
```